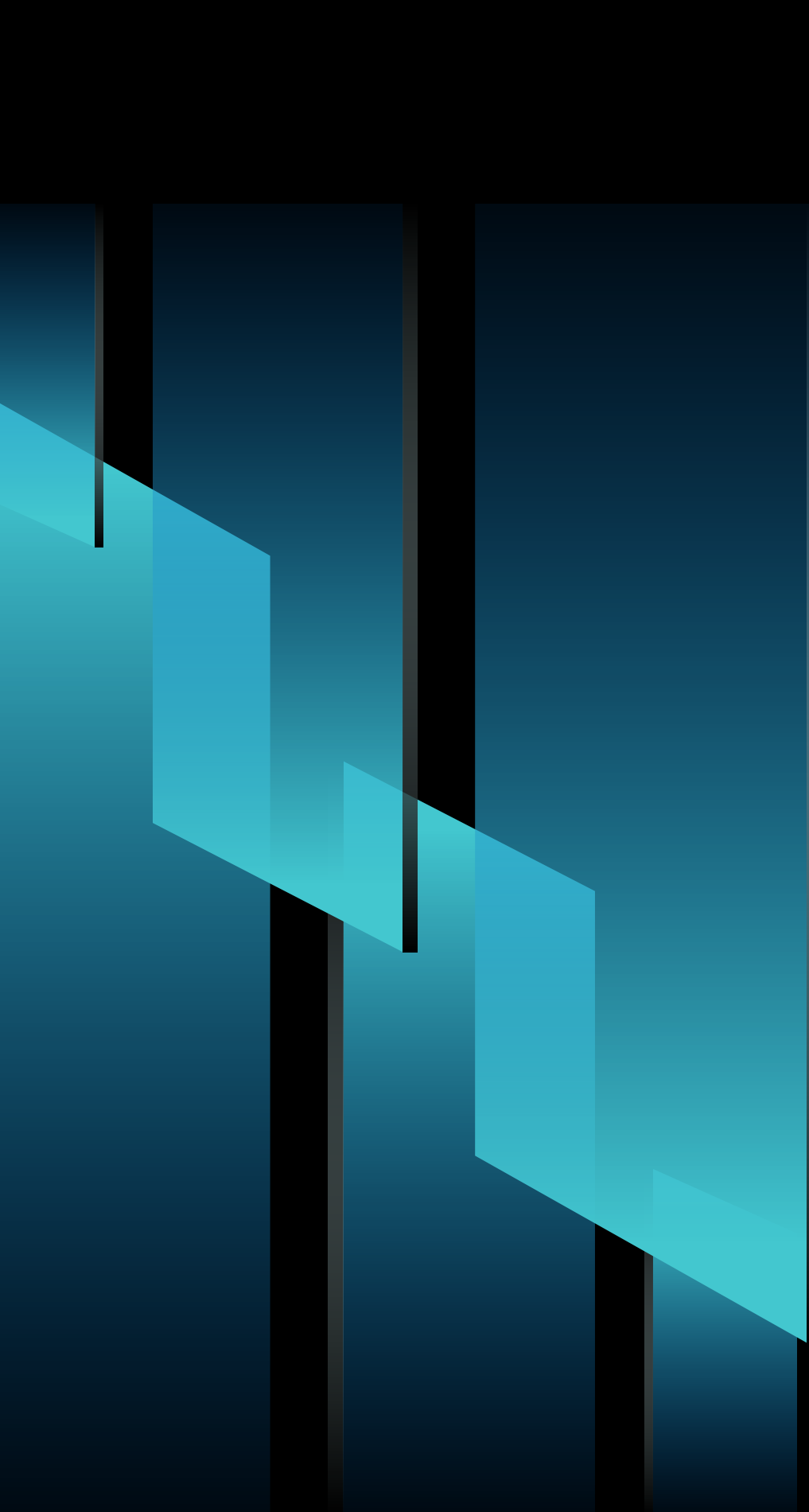




CLINIC MANAGEMENT SYSTEM

CONSOLE APP

C++ / OOP PROJECT

The logo for Cairo Coding School is a purple square with a thin white circle inside. The text "CAIRO; CODING SCHOOL" is written in yellow, bold, sans-serif capital letters, centered within the circle.

**CAIRO;
CODING
SCHOOL**

PREPARED BY :

Mohannad Reda

• Features

- ```
1. Add patient
2. View all patients
3. Search by ID
4. Sort patients
5. Delete patient
6. View patient history
0. Back
```

```
Choose: |
```

```
* PATIENT *
```

- ```
1. Add doctor
2. View all doctors
3. Search by ID
0. Back
```

```
Choose: |
```

```
* DOCTOR *
```

- ```
1. Add to waiting room
2. View who is waiting
3. Call next patient
0. Back
```

```
Choose: |
```

```
* WAITING ROOM *
```

# Inheritance

```
class Person {
protected:
 string name;
 int id;
public:
 Person(int id);
 virtual void display
 string getName();
 int getID();
};
```

```
class Patient : public Person {
private:
 string phone, age;
 static int ID;
 History history;
public:
 Patient();
 void display() override;
 string getAge();
 string getPhone();
 void addVisitToHistory(Visit* visit);
 void viewHistory();
};
```

```
class Doctor : public Person {
private:
 static int ID;
 string specialization;
 string consultant_fee;
public:
 Doctor();
 void display() override;
 string getSpecialization();
 string getFee();
};
```

# Algorithms

## Sorting Algorithm

```
int Sort::selectionByID(vector<Patient>& patient) {
 int comparisons = 0;
 int size = patient.size();
 for (int i = 0; i < size - 1; i++) {
 int minIndex = i;
 for (int j = i + 1; j < size; j++) {
 comparisons++;
 if (patient.at(j).getID() < patient.at(minIndex).getID())
 minIndex = j;
 }
 swap(patient.at(i), patient.at(minIndex));
 }
 return comparisons;
}
```

$O(n^2)$

## Searching Algorithm

```
int Search::binarySearch(vector<Patient>& patient, int target) {
 int first = 0;
 int last = patient.size() - 1;
 while (first <= last) {
 int mid = (first + last) / 2;
 if (target == patient.at(mid).getID()) return mid;
 else if (target > patient.at(mid).getID()) first = mid + 1;
 else last = mid - 1;
 }
 return -1;
}
```

$O(\log(n))$

# Data Structure

## Use Linked List To Store Patient History

```
void History::addVisit(Visit* node) {
 Node* visit = new Node(node);
 if (head == nullptr) {
 head = visit;
 return;
 }
 Node* temp = head;
 while (temp->next != nullptr) { temp = temp->next; }
 temp->next = visit;
}

void History::viewVisits() {
 Node* temp = head;
 while (temp != nullptr) {
 temp->visit->view();
 temp = temp->next;
 }
}
```

```
struct Node {
 Visit* visit;
 Node* next;
 Node(Visit* v) : visit(v), next(nullptr) {}
 ~Node() { delete visit; }
};
```

# Memory Management

Use Destructor  
to Prevent memory  
leaks and clearing the  
heap.

```
History::~~History() {
 Node* temp = head;
 while (temp != nullptr) {
 Node* node = temp->next;
 delete temp;
 temp = node;
 }
}
```

```
struct Node {
 Visit* visit;
 Node* next;
 Node(Visit* v) : visit(v), next(nullptr) {}
 ~Node() { delete visit; }
};
```

# Key Features & Live Demo

## Add Patient

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Name : Mohannad Reda

Phone : 01168944537

Age : 20

[OK] Patient 1 added.

| ID | NAME          | AGE | PHONE       |
|----|---------------|-----|-------------|
| 1  | Mohannad Reda | 20  | 01168944537 |
| 2  | Ahmed Amr     | 25  | 01067493728 |

## View Patients

# Add Appointment to WaitingRoom

## Call Next Patient

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Patient ID : 2

Doctor ID : 1

Visit Date (YYYY-MM-DD) : 2026-10-01

Diagnosis : Heart Pain

Case type : (1) Emergency (2) Normal : 1

[OK] Ahmed Amr added as Emergency

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>>> NOW CALLING: Ahmed Amr [Emergency]

Doctor : Dr. Mahmoud Ahmed

Diagnosis : Heart Pain

Fee : 450

[OK] Visit added to history. Waiting room is empty.

The background is a dark navy blue. On the left side, there are several overlapping, semi-transparent blue geometric shapes, including rectangles and parallelograms, some of which are tilted. On the right side, there is a grid of small, light blue dots that fades out towards the center.

***Thank You***